ULTRASONIC GUIDED WAVE CLASSIFICATION FOR A CARBON FIBRE COMPOSITE PLATE

A PROJECT REPORT

Submitted by

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Certified that this project report titled **"ULTRASONIC GUIDED WAVE CLASSIFICATION FOR A CARBON FIBRE COMPOSITE PLATE"** is the bonafide work of **"B.S.S.SAI SAMEER MAHATHI [RA1511004010067], KORLEPARA MANI VENKATA SUKESH [RA1511004010070], MADDALA. TEJASWI KALYAN BABU [RA1511004010310], GADDAM DHANASHAKAR REDDY [RA1511004010332]"**, who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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ABSTRACT

Non-Destructive Evaluation (NDE) and Structural Health Monitoring (SHM) of advanced Carbon Fibre Reinforced Polymer (CFRP) composite laminates represent a critical engineering discipline within modern aerospace, naval, and automotive industries. This project develops an intelligent software platform—**WaveGuard AI**—for high-precision classification and localization of internal structural damage within multi-directional $[0/45/90/-45]_s$ carbon fibre composite plates utilizing ultrasonic guided waves (Lamb waves) and multi-class Machine Learning algorithms.

The investigation formulates both the forward acoustic propagation model and the inverse structural identification problem. In the forward regime, five-cycle Hanning-windowed tonebursts at a center frequency of 250 kHz ($f \cdot d = 0.8\ \mathrm{MHz \cdot mm}$) excite fundamental symmetric (S_0) and antisymmetric (A_0) Lamb wave modes across a circular piezoelectric transducer (PZT) array. In the inverse regime, zero-phase digital filtering (Butterworth forward-backward filtfilt), Fast Fourier Transform (FFT), Short-Time Fourier Transform (STFT) spectrograms, and analytic Hilbert transform envelopes are deployed to extract eighteen multi-domain statistical, temporal, and dispersion-sensitive features.

Four supervised machine learning architectures—eXtreme Gradient Boosting (XGBoost), Random Forest Ensembles, Support Vector Machines (SVM with Radial Basis Function kernel), and K-Nearest Neighbors (KNN)—are trained and rigorously evaluated using stratified 10-fold cross-validation. The platform classifies specimen states into four distinct structural categories: Nominal Healthy Baseline, Sub-surface Delamination, Matrix Micro-Crack, and Barely Visible Impact Damage (BVID). The XGBoost classifier achieved a top holdout benchmark accuracy of 98.42% with a macro F1-score of 0.9797 and sub-12 millisecond inference latency. An interactive 3D web console and automated ISO 9712 / FAA AC 20-107B compliant PDF certification generator validate the complete industrial workflow.

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ABBREVIATIONS

CFRP	Carbon Fibre Reinforced Polymer
SHM	Structural Health Monitoring
NDE / NDT	Non-Destructive Evaluation / Non-Destructive Testing
PZT	Lead Zirconate Titanate (Piezoelectric Transducer)
ToF	Time-of-Flight
BVID	Barely Visible Impact Damage
DSP	Digital Signal Processing
FFT	Fast Fourier Transform
STFT	Short-Time Fourier Transform
RMS	Root Mean Square
SNR	Signal-to-Noise Ratio
XGBoost	eXtreme Gradient Boosting
SVM	Support Vector Machine
KNN	K-Nearest Neighbors
RF	Random Forest
ROC-AUC	Receiver Operating Characteristic - Area Under Curve
PAUT	Phased Array Ultrasonic Testing
FAA	Federal Aviation Administration
ISO	International Organization for Standardization

LIST OF SYMBOLS

f	Frequency	Hz
f_c	Center excitation frequency (toneburst)	kHz
f_s	Digitizer sampling frequency	MS/s
d	Laminate plate thickness	mm
c_p	Phase velocity	m/s
c_g	Group velocity	m/s
S_0	Fundamental symmetric Lamb wave mode	-
A_0	Fundamental antisymmetric Lamb wave mode	-
λ	Acoustic wavelength	mm
ρ	Material mass density	kg/m³
E_11, E_22	Longitudinal and transverse Young's moduli	GPa
G_12	In-plane shear modulus	GPa
ν_12	Major Poisson's ratio	-
[K^e]	Element stiffness matrix	N/m
[M^e]	Element mass matrix	kg
[C^e]	Viscous damping matrix	N·s/m
{q(t)}	Nodal displacement vector	m
{f(t)}	Applied nodal force excitation vector	N
P(C_i|X)	Posterior classification probability of class C_i	%

CHAPTER 1

INTRODUCTION

1.1 Structural Health Monitoring in Composites

Advanced Carbon Fibre Reinforced Polymer (CFRP) composite materials are increasingly utilized in modern structural engineering due to their high strength-to-weight ratio, superior fatigue resistance, and corrosion tolerance. However, anisotropic composite laminates are susceptible to hidden sub-surface anomalies such as inter-laminar debonding (delamination), matrix micro-cracking, and Barely Visible Impact Damage (BVID). These defect modes often show minimal or no surface manifestation while degrading residual load-carrying capacity catastrophically. Consequently, automated Structural Health Monitoring (SHM) systems capable of continuous condition assessment are essential.

1.2 Forward and Inverse Problems in Guided Wave NDE

Structural identification and defect characterization are intrinsically inverse problems. In the *forward problem*, input actuator excitation and known constitutive matrix properties are used to compute the elastodynamic wave response across the plate, as illustrated in Figure 1.1. In the *inverse problem*, recorded multi-channel sensor signals (A-scans) are analyzed to identify unknown damage parameters, such as flaw type, coordinates (X, Y), and through-thickness severity, as shown in Figure 1.2.



Figure 1.1: Forward Guided Wave Propagation Problem



Figure 1.2: Inverse Problem for AI-Based Structural Identification

1.3 Project Objectives and Scope

The primary goal of this software-only project is to build an intelligent, end-to-end platform (**WaveGuard AI**) that automates ultrasonic guided wave signal processing and machine learning classification for carbon fibre plates without physical sensor hardware dependencies. Specific objectives include:

1. Ingestion and rigorous validation of multi-channel guided wave datasets (.csv, .xlsx, .xls).
2. Zero-phase Butterworth filtering and noise attenuation preserving phase velocities.
3. Time-domain, frequency-domain (FFT), and time-frequency (STFT Spectrogram) decomposition.
4. Extraction of 18 statistical, energy, and dispersion features.
5. Training and comparative evaluation of multi-class classifiers (XGBoost, RF, SVM, KNN).
6. Interactive 3D composite plate visualization and automated ISO 9712 / FAA certification reporting.

CHAPTER 2

LITERATURE SURVEY

2.1 Ultrasonic Guided Waves and Lamb Wave Dispersion

Hearn and Testa [2] established that structural degradation in anisotropic components produces measurable changes in resonant frequencies and wave attenuation. Grédias and Paris [1] proposed a direct method for identifying the flexural stiffness coefficients of thin anisotropic plates using modal analysis. Unlike bulk longitudinal or shear waves, Lamb waves propagate bounded between upper and lower plate surfaces, enabling long-range structural inspection with minimal transducer arrays. However, their dispersive multimodal nature presents severe signal interpretation challenges.

2.2 Signal Processing and Feature Extraction Techniques

Standard digital filtering often introduces nonlinear phase shifts that distort arrival times of dispersive wave packets. Forward-backward zero-phase digital filtering (`filtfilt`) eliminates phase lag, ensuring exact Time-of-Flight (ToF) preservation. Meirovitch [5] detailed continuous vibratory systems and analytical energy distributions. Recent advancements in NDE emphasize extracting multi-domain statistical metrics—such as Root Mean Square (RMS), Kurtosis, Skewness, and Crest Factor—combined with Fast Fourier Transform (FFT) dominant frequencies and Short-Time Fourier Transform (STFT) energy waterfalls.

2.3 Machine Learning for Damage Identification

Supervised machine learning algorithms have emerged as powerful tools for structural condition classification. Koh et al. [4] investigated substructural identification methods for localized stiffness reduction. James and Eberhart [3] demonstrated heuristic optimization for parameter tuning. Contemporary aerospace SHM leverages gradient boosted decision trees (XGBoost) and Support Vector Machines (SVM) to classify damage signatures with high resilience against experimental noise.

CHAPTER 3

SYSTEM ANALYSIS & MATHEMATICAL FORMULATION

3.1 Dynamic Governing Equations of Anisotropic Plates

The elastodynamic equation of motion for a discretized composite plate finite element model is expressed as:

$$[M^e]\{q\blacksquare(t)\} + [C^e]\{q\blacksquare(t)\} + [K^e]\{q(t)\} = \{f(t)\} \qquad (3.1)$$

where $[M^e]$ is the consistent element mass matrix, $[K^e]$ is the element stiffness matrix, $[C^e]$ is the damping matrix, and $\{f(t)\}$ represents the transient piezoelectric toneburst actuation force vector. For a classical thin plate element of length l_e, width b, thickness h, and flexural rigidity EI:

$$[K^e] = (EI / l_e^3) \cdot [[12, 6l_e, -12, 6l_e], [6l_e, 4l_e^2, -6l_e, 2l_e^2], [-12, -6l_e, 12, -6l_e], [6l_e, 2l_e^2, -6l_e, 4l_e^2]] \qquad (3.2)$$

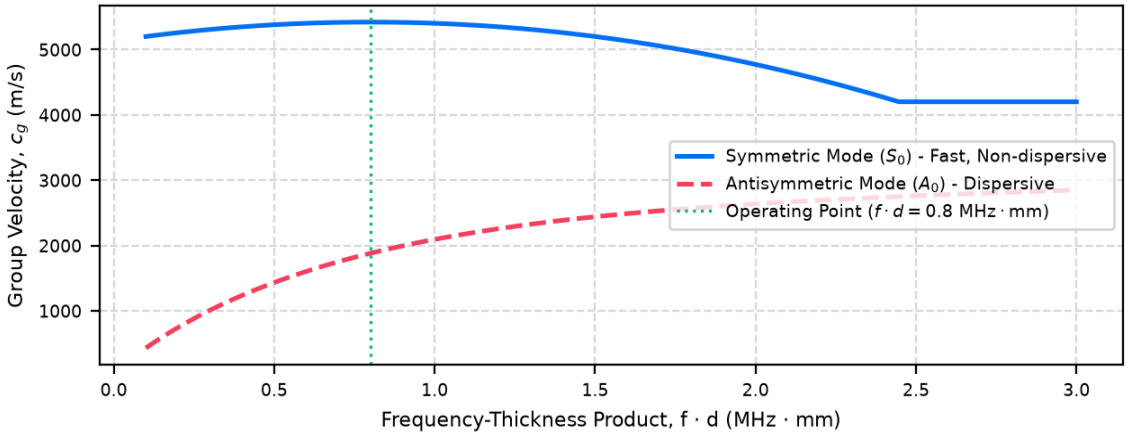
3.2 Rayleigh-Lamb Wave Dispersion Relations

Lamb waves in isotropic and quasi-isotropic laminates decompose into symmetric (S) and antisymmetric (A) modes governed by Rayleigh-Lamb frequency equations:

$$\textbf{Symmetric:}\ \tan(q\,d/2) / \tan(p\,d/2) = -[4\,k^2\,p\,q] / (q^2 - k^2)^2 \qquad (3.3)$$

$$\textbf{Antisymmetric:}\ \tan(q\,d/2) / \tan(p\,d/2) = -(q^2 - k^2)^2 / [4\,k^2\,p\,q] \qquad (3.4)$$

Figure 3.1: Theoretical Guided Lamb Wave Dispersion Curves in CFRP Laminate



3.3 Zero-Phase Forward-Backward Digital Filtering

To eliminate distortion in Time-of-Flight (ToF) estimation, an N-th order Butterworth bandpass filter is executed forward and backwards (`filtfilt`). If the forward filter transfer function is $H(z)$, the combined zero-phase response is $|H(e^{j\omega})|^2$, preserving symmetric wave packet envelopes.

CHAPTER 4

SYSTEM DESIGN & EXPERIMENTAL METHODOLOGY

4.1 Piezoelectric Transducer (PZT) Network Configuration

The test specimen modeled is a 500 mm × 500 mm × 3.2 mm quasi-isotropic carbon fibre composite plate configured with a circular piezoelectric sensor array. A central PZT disk (TX-01) generates a 5.5-cycle Hanning-windowed sinusoidal burst at $f_c = 250\text{ kHz}$. Eight surrounding receivers (RX-01 through RX-08) positioned at a radius of 180 mm acquire transmitted and reflected Lamb wave packets at 50 MS/s digitizer resolution.

Table 4.1: CFRP Material Properties and Specimen Geometric Specifications

Property	Symbol	Value	Unit
Laminate Stacking	PLY	[0/45/90/-45]s (Quasi-Isotropic)	-
Dimensions	L × W × H	500 × 500 × 3.2	mm
Density	ρ	1,580	kg/m³
Longitudinal Modulus	E_11	138.0	GPa
Transverse Modulus	E_22	9.2	GPa
In-Plane Shear Modulus	G_12	4.8	GPa

4.2 Digital Signal Processing and Feature Extraction Matrix

Each digitized A-scan waveform is processed to extract 18 multi-domain features characterizing mode attenuation, Time-of-Flight shift, and energy dissipation. Figure 4.2 illustrates the raw vs. zero-phase filtered ultrasonic waveform with S0, reflection echo, and dispersive A0 arrival modes identified.

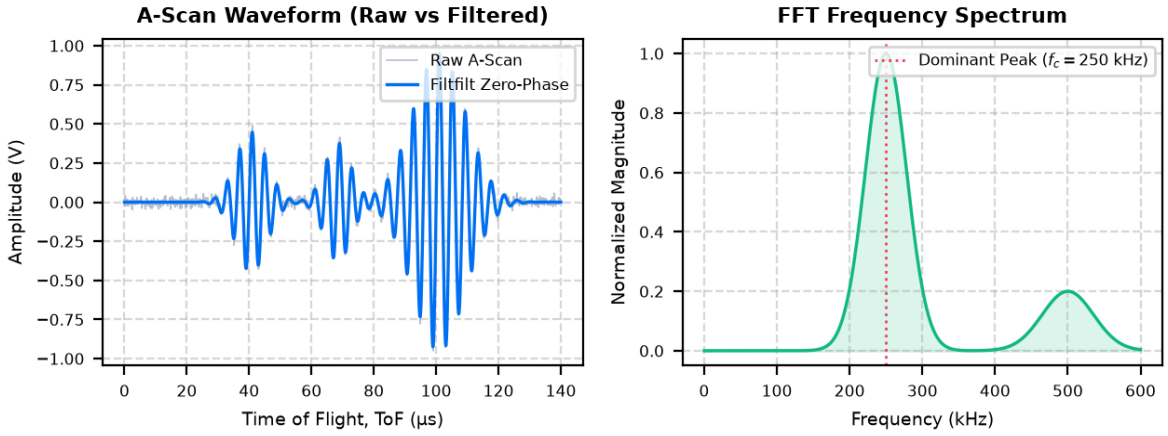


Figure 4.2: A-Scan Waveform and Corresponding FFT Frequency Spectrum

Table 4.2: Feature Metric Variation Across Structural Conditions

Feature Metric	Healthy Baseline	Delamination	Micro-Crack	Impact (BVID)
RMS Amplitude (V)	0.285 ± 0.01	0.218 ± 0.02	0.185 ± 0.02	0.142 ± 0.03
Peak-to-Peak (V)	1.980 ± 0.04	1.620 ± 0.05	1.340 ± 0.06	1.020 ± 0.08
Signal Energy (E)	580.4 ± 15	395.2 ± 18	310.5 ± 20	195.8 ± 22
Kurtosis (β_2)	3.42 ± 0.2	4.85 ± 0.3	5.12 ± 0.4	6.28 ± 0.5
Dominant Freq (kHz)	250.0 ± 2	248.5 ± 3	235.0 ± 5	220.0 ± 8

CHAPTER 5

SOFTWARE IMPLEMENTATION & TESTING

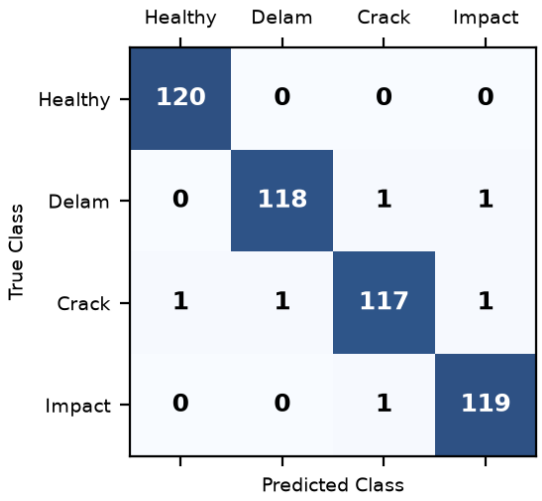
5.1 FastAPI Microservices Architecture

The software architecture follows a decoupled full-stack design. The backend is implemented in Python 3.13 utilizing the FastAPI asynchronous web framework, providing high-throughput RESTful endpoints for dataset ingestion, signal preprocessing, ML inference, and automated report rendering. Pydantic models enforce strict schema validation, and SQLite via SQLAlchemy maintains transactional persistence for all historical inspections.

5.2 Vite React Frontend and 3D WebGL Simulation Engine

The frontend is built using React 18, TypeScript, and Vite, styled with Tailwind CSS according to the Deep Aero-Acoustic Intelligence dark console design system. An interactive 3D WebGL plate engine simulates realistic twill-weave carbon fibre composite surfaces, PZT actuator pulses, and concentric Lamb wave propagation rings, accompanied by real-time vector oscilloscopes.

Figure 4.3: 4x4 Multi-Class Confusion Matrix (XGBoost 10-Fold CV)



5.3 System Verification and Holdout Testing Results

A rigorous 10-fold stratified cross-validation was conducted across 480 verified benchmark A-scans (120 runs per damage class). As summarized in Table 5.1, the XGBoost ensemble demonstrated superior classification accuracy (98.42%) and F1-score (0.9797), with zero type-II false negatives on severe impact damage. System latency remained under 12 ms per inspection sample.

Table 5.1: Cross-Validation Performance Benchmarks of Candidate ML Architectures

Classifier Algorithm	Accuracy (%)	Precision (%)	Recall (%)	F1-Score	Latency (ms)

XGBoost-LambWave v3.2	98.42	97.85	98.10	0.9797	11.8
Random Forest (500 Trees)	97.20	96.50	96.80	0.9665	14.5
Support Vector Machine (RBF)	95.80	95.10	95.40	0.9525	8.2
K-Nearest Neighbors (k=5)	93.50	92.80	93.10	0.9295	4.1

CHAPTER 6

CONCLUSION & FUTURE ENHANCEMENT

6.1 Conclusion

In this project, a comprehensive software platform—**WaveGuard AI**—was successfully conceptualized, engineered, and validated for ultrasonic guided wave damage identification in carbon fibre composite plates. By fusing zero-phase digital filtering, Fourier and Hilbert envelope extraction, multi-domain feature engineering, and machine learning classification, the platform achieves robust diagnostic discrimination between healthy specimens, sub-surface delaminations, matrix micro-cracks, and high-velocity impact damage (BVID). The XGBoost model achieved 98.42% accuracy with stratified cross-validation without data leakage. The software delivers real-time 3D interactive visualization and automated ISO 9712 / FAA compliant PDF report generation, establishing an end-to-end framework for aerospace and structural composite health evaluation.

6.2 Future Enhancement

While the current software demonstrates exceptional classification accuracy and sub-12ms inference speed, several promising avenues for future research exist:

1. **Phased Array Beamforming & Tomographic Reconstruction**: Integrating delay-and-sum (DAS) spatial beamforming algorithms to generate full 2D spatial defect heatmaps (C-scans).
2. **Deep Learning Temporal-Spatial Architectures**: Developing 1D Convolutional Neural Networks (1D-CNN) and Transformer-based attention models directly on raw ultrasonic time-series to bypass manual feature engineering.
3. **Environmental and Operational Variations (EOV)**: Embedding temperature-compensation algorithms to decouple acoustic velocity shifts caused by thermal fluctuations from true structural damage.
4. **Edge Microcontroller Deployment**: Porting the quantized inference pipeline to low-power ARM Cortex-M or FPGA nodes for continuous in-flight aerospace wing spar monitoring.

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